Joint Entropy and Conditional entropy

Joint entropy

$$H(X,Y) = \sum_{xy \in \mathcal{A}_X \mathcal{A}_Y} P(x,y) \log \frac{1}{P(x,y)}$$

Independent random variables

- Two independent probability distributions $P_X = \{p_1, \ldots, p_N\}$ & $Q_Y = \{q_1, \ldots, q_M\}$

$$H = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(i,j)\log\left(p(i,j)\right)$$

$$p(i,j) = p(i)q(j)$$

$$H = -\sum_{i=1}^{N}\sum_{j=1}^{M} p(i)q(j)\log\left(p(i)q(j)\right)$$

$$= -\sum_{i=1}^{N}\sum_{j=1}^{M} p(i)q(j)\left[\log\ p(i) + \log\ q(j)\right]$$

$$= -\sum_{i=1}^{N}\sum_{j=1}^{M} p(i)q(j)\log\ p(i) + -\sum_{i=1}^{N}\sum_{j=1}^{M} p(i)q(j)\log\ q(j)$$

$$= -\sum_{i=1}^{N} p(i)\log\ p(i)\{q_1 + \ldots + q_M\} + -\sum_{j=1}^{M} q(j)\log\ q(j)\{p_1 + \ldots + p_N\}$$

$$\{q_1 + \ldots + q_M\} = \{p_1 + \ldots + p_N\} = 1$$

$$H = -\sum_{i=1}^{N} p(i)\log\ p(i) + -\sum_{j=1}^{M} q(j)\log\ q(j)$$

$$= H_{P_X} + H_{Q_Y}$$

Conditional entropy of X given $y=b_k$

$$H(X \mid y=b_k) \equiv \sum_{x \in \mathcal{A}_X} P(x \mid y=b_k) \log \frac{1}{P(x \mid y=b_k)}$$

Conditional entropy of X given $y=b_1$

$$H(X \mid y=b_1) \equiv \sum_{x \in \mathcal{A}_X} P(x \mid y=b_1) \log \frac{1}{P(x \mid y=b_1)}$$

Conditional entropy of X given $y=b_2$

$$H(X \mid y=b_2) \equiv \sum_{x \in \mathcal{A}_X} P(x \mid y=b_2) \log \frac{1}{P(x \mid y=b_2)}$$

Conditional entropy of X given $y=b_3$

$$H(X \mid y=b_3) \equiv \sum_{x \in \mathcal{A}_X} P(x \mid y=b_3) \log \frac{1}{P(x \mid y=b_3)}$$

Conditional entropy of X given Y (for various y s)

$$P(y=b_1).\sum_{x\in\mathcal{A}_X} P(x\,|\,y=b_1)\log\frac{1}{P(x\,|\,y=b_1)}$$

$$+\ P(y=b_2).\sum_{x\in\mathcal{A}_X} P(x\,|\,y=b_2)\log\frac{1}{P(x\,|\,y=b_2)}$$

$$+\ P(y=b_3).\sum_{x\in\mathcal{A}_X} P(x\,|\,y=b_3)\log\frac{1}{P(x\,|\,y=b_3)}$$

$$+\ \ldots$$

Recall, $P(y).P(x|y) = P(x,y)$

Conditional entropy of X given Y (i.e. all possible y s)

$$\sum_{xy \in \mathcal{A}_X \mathcal{A}_Y} P(x, y) \log \frac{1}{P(x \mid y)}$$

Example – Find H(X,Y), H(X), H(Y), & H(X|Y)

$P(x,y)$		x			
		1	2	3	4
	1	$1/8$	$1/16$	$1/32$	$1/32$
y	2	$1/16$	$1/8$	$1/32$	$1/32$
	3	$1/16$	$1/16$	$1/16$	$1/16$
	4	$1/4$	0	0	0

Solution for H(X,Y)

Find out information content

	1	2	3	4
1	1/8	1/16	1/32	1/32
2	1/16	1/8	1/32	1/32
3	1/16	1/16	1/16	1/16
4	1/4	0	0	0

	1	2	3	4
1	3	4	5	5
2	4	3	5	5
3	4	4	4	4
4	2	0	0	0

Find out average information content

	1	2	3	4
1	1/8	1/16	1/32	1/32
2	1/16	1/8	1/32	1/32
3	1/16	1/16	1/16	1/16
4	1/4	0	0	0

X

	1	2	3	4
1	3	4	5	5
2	4	3	5	5
3	4	4	4	4
4	2	0	0	0

	1	2	3	4
1	3/8	4/16	5/32	5/32
2	4/16	3/8	5/32	5/32
3	4/16	4/16	4/16	4/16
4	2/4	0	0	0

Find out average information content

	1	2	3	4
1	3/8	4/16	5/32	5/32
2	4/16	3/8	5/32	5/32
3	4/16	4/16	4/16	4/16
4	2/4	0	0	0

Sum up

108/32 = 27/8 = 3.4 bits

Solution for H(X) and H(Y)

$P(x,y)$		x				$P(y)$	
		1	2	3	4		**I(y)**
y	1	$1/8$	$1/16$	$1/32$	$1/32$	$1/4$	2
	2	$1/16$	$1/8$	$1/32$	$1/32$	$1/4$	2
	3	$1/16$	$1/16$	$1/16$	$1/16$	$1/4$	2
	4	$1/4$	0	0	0	$1/4$	2
$P(x)$		$1/2$	$1/4$	$1/8$	$1/8$		

I(x)	**1**	**2**	**3**	**3**

Find average of H(X)

$P(x)$	$1/2$	$1/4$	$1/8$	$1/8$

X

I(x)	**1**	**2**	**3**	**3**
=	1/2	2/4	3/8	3/8

Sum up

14/8 = 7/4 = 1.75 bits

Find average of H(Y)

$P(y)$

$1/4$
$1/4$
$1/4$
$1/4$

X

I(y)	=
2	2/4
2	2/4
2	2/4
2	2/4

Sum up
2 bits

Solution for H(X|Y)

Recall, $P(x,y) = P(x|y).P(y)$

Solution for H(X|Y)

$P(x,y)$		x				$P(y)$
		1	2	3	4	
y	1	$1/8$	$1/16$	$1/32$	$1/32$	$1/4$
	2	$1/16$	$1/8$	$1/32$	$1/32$	$1/4$
	3	$1/16$	$1/16$	$1/16$	$1/16$	$1/4$
	4	$1/4$	0	0	0	$1/4$
$P(x)$		$1/2$	$1/4$	$1/8$	$1/8$	

$P(x \mid y)$		x			
		1	2	3	4
y	1	$1/2$	$1/4$	$1/8$	$1/8$
	2	$1/4$	$1/2$	$1/8$	$1/8$
	3	$1/4$	$1/4$	$1/4$	$1/4$
	4	1	0	0	0

Find out information content of (X|Y)

	1	2	3	4
1	1/2	1/4	1/8	1/8
2	1/4	1/2	1/8	1/8
3	1/4	1/4	1/4	1/4
4	1	0	0	0

	1	2	3	4
1	1	2	3	3
2	2	1	3	3
3	2	2	2	2
4	0	0	0	0

Find out average information content

	1	2	3	4
1	1/2	1/4	1/8	1/8
2	1/4	1/2	1/8	1/8
3	1/4	1/4	1/4	1/4
4	1	0	0	0

X

	1	2	3	4
1	1	2	3	3
2	2	1	3	3
3	2	2	2	2
4	0	0	0	0

	1	2	3	4
1	1/2	2/4	3/8	3/8
2	2/4	1/2	3/8	3/8
3	2/4	2/4	2/4	2/4
4	0	0	0	0

	1	2	3	4	(sum-up row wise) H(X|Y)	
1	1/2	2/4	3/8	3/8	14/8	← H(X|Y=1)
2	2/4	1/2	3/8	3/8	14/8	← H(X|Y=2)
3	2/4	2/4	2/4	2/4	2	← H(X|Y=3)
4	0	0	0	0	0	← H(X|Y=4)

- H(X) = 1.75 bits
- H(X|Y=4) = 0 & is less than H(X) - **learning decreases entropy**
- H(X|Y=3) = 2 & is greater than H(X) - **learning increases entropy**
- H(X|Y=2) = 1.75 & is equal to H(X) - **learning doesn't change entropy**
- H(X|Y=1) = 1.75 & is equal to H(X) - **learning doesn't change entropy**

	1	2	3	4	(sum-up row wise) H(X|Y)
1	1/2	2/4	3/8	3/8	14/8
2	2/4	1/2	3/8	3/8	14/8
3	2/4	2/4	2/4	2/4	2
4	0	0	0	0	0

Find average to get H(X|Y):

=P(Y=1). H(X|Y=1)+P(Y=2). H(X|Y=2)

+P(Y=3). H(X|Y=3)+P(Y=4). H(X|Y=4)

=(14/8).(1/4) + (14/8).(1/4) + 2.(1/4)+0

=14/32+14/32+2/4 = 44/32 = 11/8 = 1.37 bits

On an average, H(Y|X) < H(X) i.e. 1.37 < 1.75